PORTABLE PLANE
Brazilian pioneer Alberto Santos-Dumont's tiny 19 Demoiselle monoplane, built in 1907, had a wingspan of just 18ft (6m) and was perhaps the first ultralight. It was designed as an aerial "runabout" and easily separated into two parts (the tail, and the wings and propeller), to allow for easy transportation.

brothers' flights in 1903 deserved to be called "the first," there could be no doubt whatsoever that by the end of 1905 they were the only people in the world with a practical flying machine. At this time, the brothers took the extraordinary decision to cease all further flying experiments, devoting much of their effort to a search for lucrative business contracts. The obvious potential customer for the new flying machine was the army.

The brothers suggested in a letter to their congressman, Robert Nevin, in January 1905, that the machine could be used for "scouting and carrying messages in time of war." But when Nevin raised the matter with the US War Department, the official response was dismissive. Faced with rejection at home, the Wrights approached the British and French military establishments. A French delegation visited Dayton to negotiate with the Wrights in the spring of 1906, but no agreement was reached. The crux of the problem was that the Wrights would not demonstrate their flying machine until someone had signed a contract to buy it, but potential buyers were reluctant to commit without seeing the machine in action.

The Wrights' decision to stop flying was extremely risky. Details of most aspects of their work were known to aviation enthusiasts. Other experimenters had a serious chance of catching up with or overtaking them. In 1907, the inventor Alexander Graham Bell set up the Aerial Experiment Association in Hammondsport, New York, bringing together a talented team – including motorcycle manufacturer Glenn H.

Curtiss – with the avowed goal of building "a practical airplane which will carry a man and be driven through the air on its own power."

Main competition

The most potent challenge to the Wright brothers came from France. In a tradition dating back to the Montgolfier brothers, the French considered themselves the natural leaders in world aviation. Reports of the Wright brothers' achievements greatly perturbed the French enthusiasts centered around the prestigious Aéro-Club de France. Some reacted by disparaging what the Wrights had done; all felt that it was their patriotic duty to prove that the French could do better. Fortunately for the Americans, France's would-be aeronauts had more

GLENN H. CURTISS

One of the founding fathers of American aviation, Glenn H. Curtiss (1878–1930) was born in Hammondsport, New York. Like the Wright brothers, Curtiss started in the bicycle business, before moving on to building and racing motorcycles. His skill in producing lightweight motorcycle engines attracted the attention of inventor Alexander Graham Bell, who in 1907 invited Curtiss to join his Aerial Experiment Association – where he played a leading part in designing a series of aircraft controlled by ailerons, rather than the wing warping used by the Wrights. On Independence Day 1908, Curtiss made the first public flight in the United States in *June Bug*.

A fearless pilot, Curtiss was often found at early aviation meetings, specializing in speed events. He eventually set up his own aircraft manufacturing company, pioneering seaplane and flying boat designs. By 1914, he was the leading aircraft manufacturer in the US.

RACING MAN
Before turning to flying in 1907, Glenn Curtiss was a successful racing motorcyclist. After winning many prizes for his flying skills, Curtiss went on to organize his own flying displays (right). His new career was hampered by a bitterly contested patent dispute with the Wright brothers over their wing-warping technology.